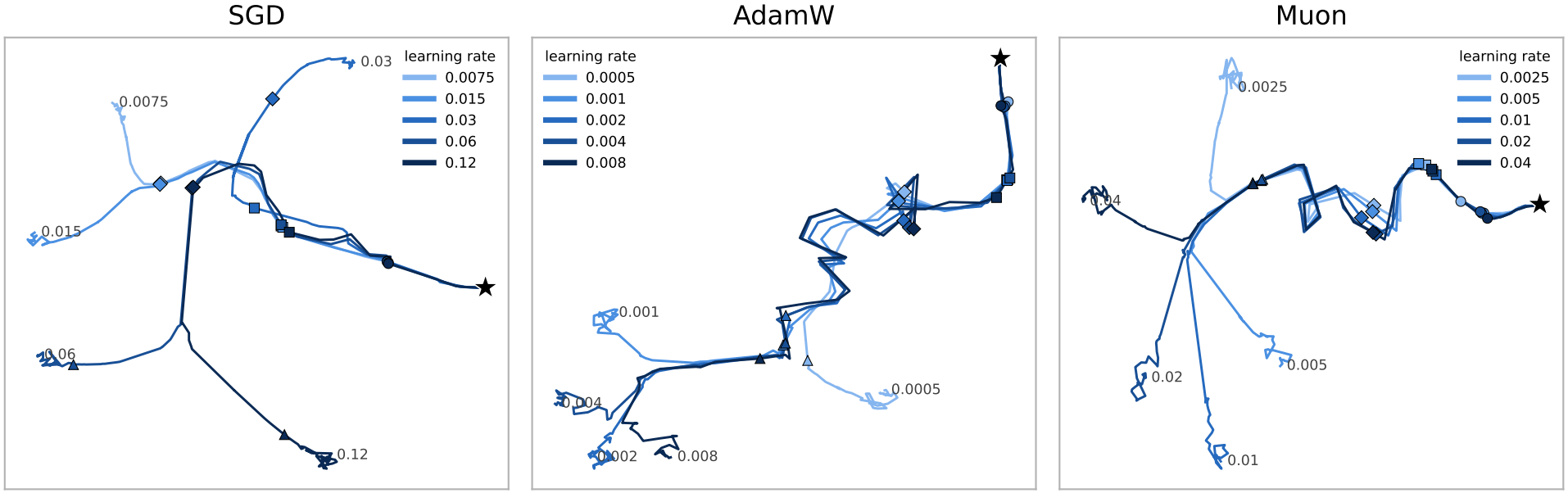
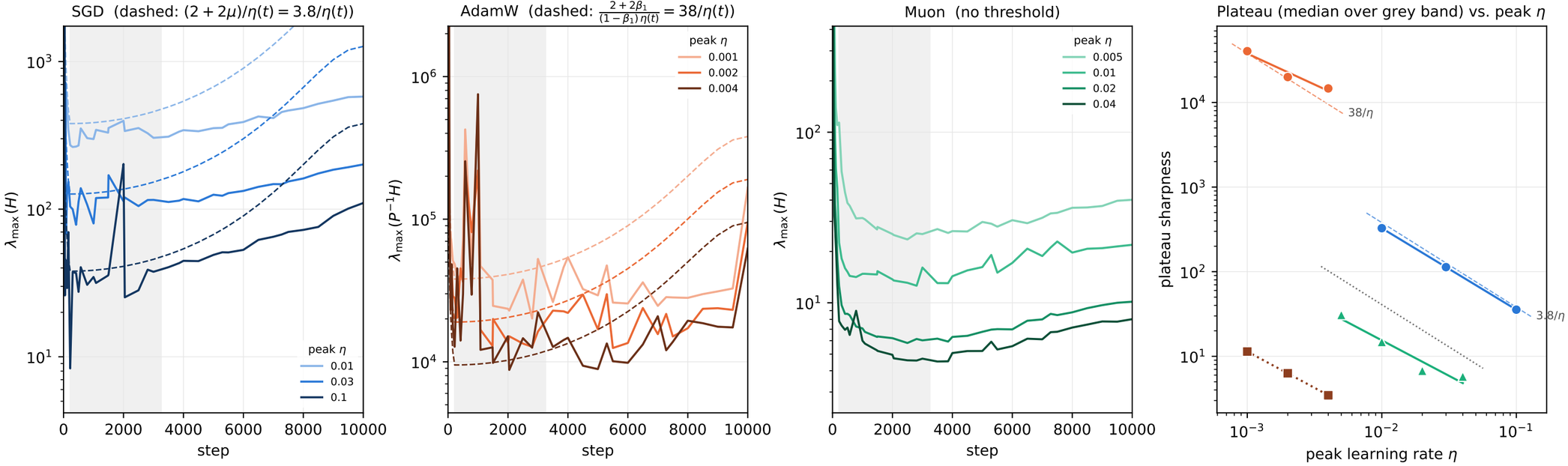
Setting: TinyStories-10M, as in trajectories_methods.pdf, data order d0 only, shared initialization; learning rates $\frac{1}{4}$, $\frac{1}{2}$, 1, 2, 4 × each optimizer's main rate: SGD 0.0075–0.12, AdamW 5×10^{-4}–8×10^{-3}, Muon 0.0025–0.04; 10,000 steps; the 1× run is the main run. densMAP of the square-root Jensen–Shannon distance, one embedding per optimizer.



Setting: TinyStories-10M, 10,000 steps, linear warmup 200 then cosine to 10%, data order 0, shared initialization. Sharpness: $\lambda_{\max}(H)$ for SGD and Muon, $\lambda_{\max}(P^{-1}H)$ for AdamW.

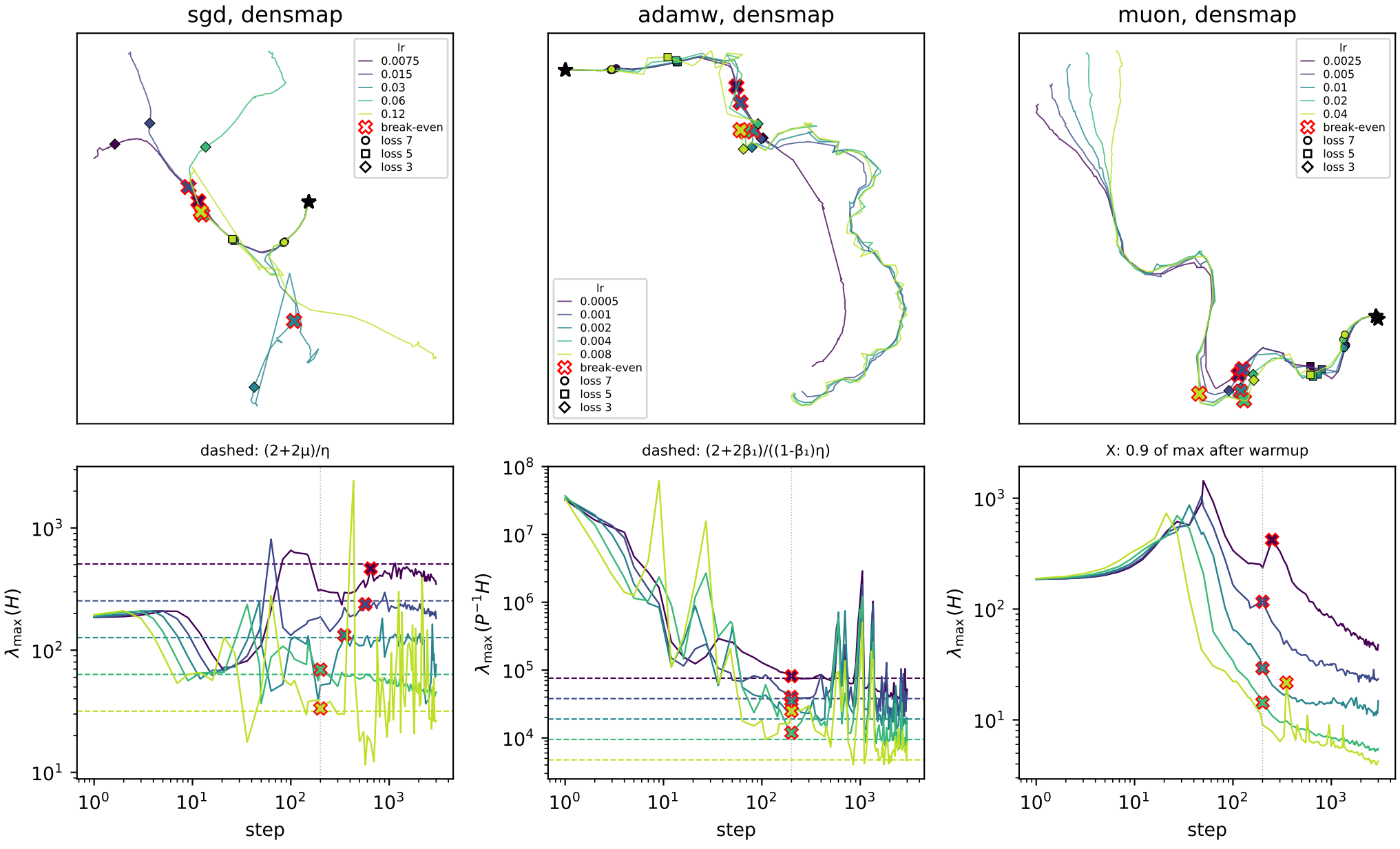
Solid: sharpness of each run; dashed, same colour: its stability threshold at the instantaneous rate $\eta(t) = \eta\,\varphi(t)$; grey band: plateau window (after warmup, $\varphi(t) \geq 0.8$).



log-log least-squares fits

- SGD $\lambda_{\max}(H)$: slope −0.96 (3 pts)
- AdamW $\lambda_{\max}(P^{-1}H)$: slope −0.73 (3 pts)
- AdamW $\lambda_{\max}(H)$: slope −0.86 (3 pts)
- Muon $\lambda_{\max}(H)$: slope −0.84 (4 pts)
- slope −1 reference
- thresholds at peak η (slope −1)

Break-even point. TinyStories-10M, suite `ts10m_breakeven_lr`: constant learning rate after a 200-step warmup, 3,000 steps, data order 0, shared initialization, five learning rates per optimizer. Top: densMAP of the square-root Jensen–Shannon distance, one embedding per optimizer; red X: break-even step. Bottom: sharpness vs. step. Break-even is the first step after warmup at which the sharpness reaches 0.9× its stability threshold (dashed; SGD $(2+2\mu)/\eta$, AdamW $(2+2\beta_1)/((1-\beta_1)\eta)$), or, for Muon (no threshold), 0.9× its maximum after warmup.



Muon spectral sharpness S vs. its stability bound $2\|\nabla L\|_*/\eta$: `ts10m_lrsweep_hessian`, step 10000 only ($\eta = 10\%$ of peak).

